

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

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Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. A software company releases an online banking system. After deployment, several transaction errors are reported by customers. Developers fix these defects without adding new functionality. Which type of software maintenance is being performed?

- A. Adaptive Maintenance
- B. Perfective Maintenance
- C. Preventive Maintenance
- D. Corrective Maintenance

Answer: D

2. A government payroll system has fixed and well-defined requirements before development begins. Each development phase is completed before moving to the next phase. Which SDLC model is the most appropriate?

- A. Agile Model
- B. Prototype Model
- C. Waterfall Model
- D. Scrum

Answer: C

3. A university develops an early version of an online admission portal to gather student feedback before building the final system. Which software process model best fits this scenario?

- A. Incremental Model
- B. Prototype Model
- C. V-Model
- D. RAD

Answer: B

4. A medical device software project requires every development phase to have a corresponding testing activity because patient safety is critical. Which SDLC model should be selected?

- A. Waterfall Model
- B. Agile Model
- C. Prototype Model
- D. V-Model

Answer: D

5. According to Boehm's Law, the cost of fixing a software defect is highest when the defect is discovered during—

- A. Requirements Analysis
- B. Design Phase
- C. Coding Phase

D. After Deployment

Answer: D

6. A banking application requires all customer passwords to be encrypted using AES-256 encryption. This requirement is classified as—

- A. Functional Requirement
- B. Business Requirement
- C. Non-functional Requirement
- D. User Requirement

Answer: C

7. Which Requirements Engineering activity ensures that documented requirements accurately represent stakeholder expectations?

- A. Elicitation
- B. Validation
- C. Specification
- D. Feasibility Study

Answer: B

8. Developers misunderstand customer expectations because stakeholders provide unclear and incomplete requirements. Which Requirements Engineering challenge does this describe?

- A. Scope Creep
- B. Communication Gap
- C. Risk Analysis
- D. Timeboxing

Answer: B

9. A software startup releases working software every two weeks and continuously incorporates customer feedback into future releases. Which development methodology is being followed?

- A. Waterfall
- B. Agile
- C. V-Model
- D. Spiral

Answer: B

10. Which Agile principle emphasizes delivering working software frequently rather than producing excessive documentation?

- A. Continuous Planning
- B. Customer Satisfaction
- C. Working Software
- D. Software Reuse

Answer: C

11. In Extreme Programming (XP), two developers work together at the same workstation to improve software quality and knowledge sharing. This practice is known as—

- A. Daily Scrum

- B. Continuous Integration
- C. Pair Programming
- D. Code Inspection

Answer: C

12. Which XP value encourages developers to openly discuss problems and share ideas throughout software development?

- A. Respect
- B. Communication
- C. Courage
- D. Simplicity

Answer: B

13. Who is responsible for maximizing the business value of a product by managing and prioritizing the Product Backlog in Scrum?

- A. Scrum Master
- B. Product Owner
- C. Development Team
- D. Project Manager

Answer: B

14. Which Scrum event is held every day to synchronize team activities and identify development obstacles?

- A. Sprint Review
- B. Sprint Retrospective
- C. Sprint Planning
- D. Daily Scrum

Answer: D

15. A software project must be completed within a fixed deadline, while less important features can be postponed to future releases. Which DSDM technique is primarily applied?

- A. Pair Programming
- B. Timeboxing
- C. Code Inspection
- D. Continuous Integration

Answer: B

16. According to the MoSCoW prioritization technique, which category represents features that are absolutely essential for successful system delivery?

- A. Could Have
- B. Won't Have
- C. Must Have
- D. Should Have

Answer: C

17. Which Feature Driven Development (FDD) process is responsible for identifying and organizing all client-valued software features before implementation?

- A. Build by Feature
- B. Design by Feature
- C. Plan by Feature
- D. Build Feature List

Answer: D

18. Which activity in Feature Driven Development helps improve software quality by systematically examining source code before release?

- A. Pair Programming
- B. Sprint Review
- C. Code Inspection
- D. Daily Scrum

Answer: C

19. Which statement best represents a common software engineering myth held by customers?

- A. Software must always follow Agile development.
- B. Once the software works, no further maintenance is required.
- C. A general statement of objectives is sufficient to begin software development.
- D. Testing is more important than requirements analysis.

Answer: C

20. An organization experiences repeated project failures because developers continuously modify project scope without formal approval. This problem is commonly known as—

- A. Risk Management
- B. Scope Creep
- C. Configuration Management
- D. Verification

Answer: B